WNDRVR
SUPPORT NETWORK

VXWORKS 7 SSL PROGRAMMER'S GUIDE, SR0650

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1. INTRODUCTION

VxWorks® 7 SSL provides native Secure Sockets Layer (SSL), Transport Layer Security (TLS), and Datagram Transport Layer Security (DTLS) services.

SSL and TLS enable secure data exchange between TCP-based network applications. DTLS enables UDP-based network applications to exchange data securely.

Note: The term SSL is used in this guide to refer to both SSL and TLS.

VxWorks and OpenSSL on page 1 Wind River SSL implements the latest OpenSSL 1.1*x* release available at the time of the VxWorks release.
SSL Cipher Suites on page 2 VxWorks SSL employs cipher suites. These suites are a combination of key exchange, server authentication, symmetric cipher, and hash algorithms.
SSL Technology Overview on page 3 The SSL protocol provides privacy and data security in transmissions between two applications over TCP/IP protocols.
Directory Structure on page 4 Wind River SSL is installed at *installDir* **/vxworks-7/pkgs_v2/security/openssl-**x_x_x .
Additional Documentation on page 4 Numerous additional sources of documentation and background reading about SSL and related technologies are available.

1. VxWorks and OpenSSL

Wind River SSL implements the latest OpenSSL 1.1*x* release available at the time of the VxWorks release.

In this implementation, VxWorks SSL uses the SSL v2, SSL v3, TLS v1, TLS v1.1, TLS v1.2, TLS v1.3, and DTLS v1 protocols, as specified in the RFCs listed in Additonal Documentation on page 5.

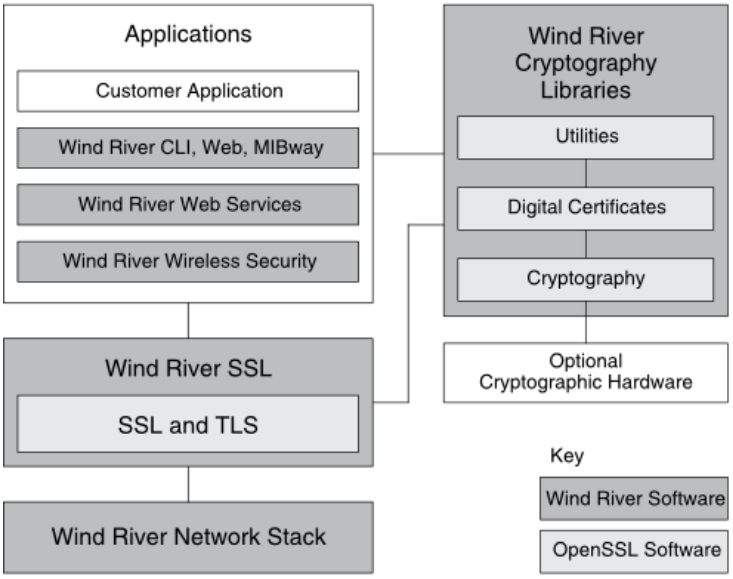
The open source SSL toolkit is available at http://www.openssl.org.

The VxWorks port of OpenSSL involves two Wind River products: VxWorks SSL and VxWorks Cryptography Libraries. To use the security services of VxWorks SSL in your application, you must use the APIs provided in VxWorks SSL and the VxWorks Cryptography Libraries, as follows:

- VxWorks SSL contains the main SSL, TLS, and DTLS functionality.
- The Cryptography Libraries implement OpenSSL. This run-time facility contains the cryptography interface (EVP), the cryptography functionality, the X.509 digital certificate functionality, and supporting utilities. For details, see the VxWorks 7 Cryptography Libraries Programmer's Guide.

The following figure summarizes the relationship between these two Wind River products, other applications, and the VxWorks Network Stack.

Figure 1. Wind River Product Relationships



OpenSSL Implementation in VxWorks

By default, the **OPENSSL** layer is enabled in VxWorks source build (VSB) projects. When you configure and build a VSB with this layer, the OpenSSL source files are included in the project. As of VxWorks 7 SR0640, you do not need to manually obtain the OpenSSL source files if you are on a system without Internet connectivity. The required OpenSSL source files are present in the following directory at installation of VxWorks 7:

installDir **/download**

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2. SSL Cipher Suites

VxWorks SSL employs cipher suites. These suites are a combination of key exchange, server authentication, symmetric cipher, and hash algorithms.

At runtime, you can list the cipher suites that are supported by the VxWorks image from the kernel shell. To do so, VxWorks must be configured with the **INCLUDE_IPSSL_USE_CMDS** component (which requires the **OPENSSL** layer in the VSB project).

Use the ciphers command from the kernel shell with the command interpreter. Use ciphers ALL to display all the suites, including the ADH suites. The -h option lists the command options.

For detailed information about the ciphers command, see the reference entry (man page) on http://www.openssl.org.

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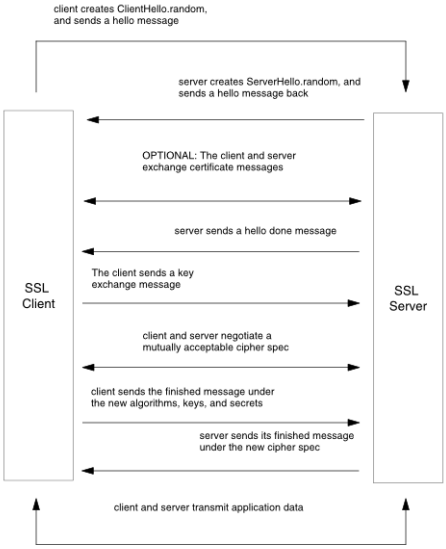
3. SSL Technology Overview

The SSL protocol provides privacy and data security in transmissions between two applications over TCP/IP protocols.

SSL allows you to use symmetric key encryption (such as RC2, RC4, DES, 3DES, or AES), public key encryption (such as DSA or RSA), cryptographic hash algorithms (such as MD5 or SHA), secure key agreement protocols (Diffie-Hellman) and message authentication codes (HMAC) to secure your data. The figure below diagrams the interaction between a client and a server to establish a secure connection using the SSL/TLS communication methodology.

Figure 2. Establishing a connection between SSL Client and Server



The steps involved in setting up a connection are:

1. The client and server exchange hello messages.
    1. The client generates **ClientHello.random**.
    2. The server generates **ServerHello.random**.
    3. The client and server exchange the following:
        - Protocol Version
        - Session ID
        - Cipher Suite
        - Compression Method
        - random numbers (**ClientHello.random** and **ServerHello.random**).
2. The client and server optionally exchange certificate information and agree upon a shared secret.
    1. The server sends the server certificate to the client. If the server has no certificate, it sends a key exchange message to the client. If the server is authenticated, it may request a certificate from the client, if that is appropriate to the cipher suite selected.
    2. The server sends the **server hello done** message, indicating that the hello-message phase of the handshake is complete. The server waits for a client response.
    3. If the server has sent a certificate request message, the client must send the certificate message.
3. The client and server exchange **key exchange** messages.
    1. The client sends a key exchange message, the content of which depends upon the public key algorithm selected between the client and server.

   2. If the client has sent a certificate with signing ability, the client sends a digitally-signed certificate verify message to the server in order to explicitly verify the certificate.
4. The client and server exchange **change cipher** messages until they agree upon a cipher spec.
   1. The client sends a change cipher spec message, and the client copies the pending Cipher Spec into the current Cipher Spec.
   2. The client sends the finished message under the new algorithms, keys, and secrets.
   3. In response, the server sends its own change cipher spec message, transfers the pending to the current Cipher Spec, and send its finished message under the new Cipher Spec.
5. When client and server have successfully negotiated the cryptography algorithm options, the actual data transmission begins.

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4. Directory Structure

Wind River SSL is installed at *installDir* **/vxworks-7/pkgs_v2/security/openssl-***x_x_x* .

The directory structure follows the format described in the following table.

Table 1. Wind River SSL Directory Structure

Directory	Contents
cdf	Configuration information that specifies what is in the SSL components
cfg	Code that configures the product and registers shell commands at startup
h	Header files

The run-time configuration routine ipssl_configure( ) is located in:

installDir **/vxworks-7/pkgs_v2/security/openssl-1.1.***x* **/cfg/ipssl_config.c**

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5. Additional Documentation

Numerous additional sources of documentation and background reading about SSL and related technologies are available.

The following sections describe additional documentation about the technologies related to Wind River SSL.

Wind River Documentation

The following Wind River document presents information associated with Wind River SSL:

VxWorks 7 Cryptography Libraries Programmer's Guide describes how to use the Wind River implementation of the cryptographic portions of OpenSSL

OpenSSL Documentation

The following documents provide a complete description of OpenSSL, how to write OpenSSL applications, and other OpenSSL topics not covered in this guide:

- Viega, J., M. Messier, and P. Chandra. *Network Security with OpenSSL*. O'Reilly Media, Inc., 2002.
  http://www.oreilly.com
- The OpenSSL Project
  http://www.openssl.org

The OpenSSL Project website also lists API reference documentation.

Other Documentation

The following documents provide more information on the SSL, TLS, and DTLS protocols:

- Freier, A., P. Karlton, and P. Kocher. *The SSL Protocol Version 3.0*. Internet Draft, March 1996.
  http://tools.ietf.org/html/draft-ietf-tls-ssl-version3-00
- Modadugu, N., and E. Rescorla. *The Design and Implementation of Datagram TLS*. Proceedings of NDSS 2004, February 2004.
  http://crypto.stanford.edu/~nagendra/papers/dtls.pdf
- Rescorla, E. *SSL and TLS: Designing and Building Secure Systems*. Addison-Wesley, 2001.
- *SSL 2.0 Protocol Specification*. November 1994.

RFCs

The following list includes IETF RFCs relevant to Wind River SSL. These RFCs can be found at: http://www.ietf.org.

- RFC 2246: *The TLS Protocol Version 1.0*
- RFC 3268: *Advanced Encryption Standard (AES) Ciphersuites for Transport Layer Security (TLS)*
- RFC 4347: *Datagram Transport Layer Security*
- RFC 4346: *The Transport Layer Security (TLS) Protocol Version 1.1*
- RFC 5246: *The Transport Layer Security (TLS) Protocol Version 1.2*
- RFC 8446: *The Transport Layer Security (TLS) Protocol Version 1.3*

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2. VXWORKS CONFIGURATION WITH SSL

VxWorks must be configured with support for SSL in a VSB and VIP project.

VSB Configuration

Layer or Option	Description
OPENSSL	Enables the OpenSSL library This layer is enabled by default.
OPENSSL_OPTIONS	Implements native OpenSSL configuration options
AUTO_FP_TASK	Enables OpenSSL assembler optimization This option is enabled by default.

The **OPENSSL_OPTIONS** option allows you to specify the following OpenSSL configuration options:

```
no-asm no-autoerrinit no-capieng no-cms no-ct no-dynamic-engine no-err no-multiblock no-nextproton
eg no-ocsp
no-psk no-rfc3779 no-sctp no-srtp no-ts no-tests no-ssl no-weak-ssl-ciphers no-tls no-dgram no-dtl
s no-aria no-blake2
no-camellia no-chacha no-cmac no-ec_nistp_64_gcc_128 no-md2 no-md4 no-mdc2 no-ocb no-poly1305 no-r
c2 no-rc4 no-rc5
no-rmd160 no-scrypt no-seed no-siphash no-sm2 no-sm3 no-sm4 no-whirlpool
```

For more information, see https://github.com/openssl/openssl/blob/master/INSTALL.md.

Using options that are not listed above may result in undefined behavior.

The **AUTO_FP_TASK** option ensures that the **VX_FP_TASK** task option is automatically added when a kernel-space or user-space task is created. The **VX_FP_TASK** task option can make context switching save and restore floating-point registers and single instruction, multiple data (SIMD) registers. The OpenSSL assembler uses SIMD registers.

If a kernel task does not use float or non-scalar operations and does not implicitly invoke OpenSSL functions, you can add the **VX_NO_FP_TASK** task option to reduce latency when the kernel is created. This option does not affect RTP tasks.

VIP Configuration

Component	Description
INCLUDE_IPSSL	Provides SSL and loads the ciphers
INCLUDE_IPSSL_USE_CMDS	Provides the command-interpreter command ciphers: **s_client**, **s_server**, **s_time**, **ssl_clt**, and **ssl_srv**
INCLUDE_IPSSL_USE_TEST_CMDS	Provides the command-interpreter command, ssltest.

3. VXWORKS SSL APPLICATIONS

VxWorks applications that use SSL can be implemented as kernel-mode or user-mode (RTP) applications.

Designing and implementing a secure application using SSL requires a thorough knowledge of security principles and how they apply to your particular security needs. It is beyond the scope of this guide to provide that level of detail. See Additional Documentation on page 4.

SSL Shared Library Use

To use SSL shared libraries with an RTP application, the application must be built as a dynamic executable.

- For Workbench:
  In the Project Explorer view, expand the RTP project and **Build Targets**.
  Right-click on the RTP executable name and select **Linker - Options**
  In the LLVM Linker Options dialog box, select **Create a dynamic executable** and click **OK**.
- For wrtool, add the -Xdynamic compiler option for Diab or the -non-static for LLVM.

The RTP application build must also use the required linkage options.

To use OpenSSL in an RTP, add the following libraries:

```
-lcrypto -lssl -lnet
```

To use OpenSSL native test APIs (for example, bftest( ) or dsatest( )) in an RTP, add the following libraries:

```
-lcrypto -lssl -lnet -lopenssltest
```

To use OpenSSL native application APIs (for example, those in the **openssl/apps** directory) in an RTP, add the following libraries:

```
-lcrypto -lssl -lnet -lapps
```

- For Workbench:
  Right-click the RTP project and select Properties.
  In the Properties dialog box, select **Build Properties > Libraries**.
  Add the required linker options to the list of library directives.
- For wrtool, add the required -l linker options.

OpenSSL Initialization

In VxWorks 7, OpenSSL initialization is handled automatically. Once you configure VxWorks with OpenSSL, as described in VxWorks Configuration With SSL, no extra steps are required to initialize OpenSSL in an application.

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